

BASIC CHEMISTRY

Module map

- Module 02: Basic Chemistry
- Connected lab: lab-02_probability-statistics.md
- Use this deck as the visual path through the module's evidence and retrieval work.

Module 02

Basic Chemistry

1

Atoms and
elements

2

Bonding
and
molecular
shape

3

Water as a
biological
solvent

4

pH and
chemical
balance

5

Chemical
reactions in
cells**Lab evidence**

lab-02_probability-statistics.md

Teaching note: Open by naming the module claim and pointing students to the lab as the evidence surface.

BASIC CHEMISTRY

Learning objectives

- Use atomic structure to explain basic bonding patterns.
- Distinguish covalent, ionic, and hydrogen bonding in biological examples.
- Explain why water properties matter for cells and organisms.
- Interpret pH as a chemical condition that affects biology.
- Track atoms through a simple chemical reaction.

OBJECTIVE 1

Use atomic structure to explain basic bonding patterns.

OBJECTIVE 2

Distinguish covalent, ionic, and hydrogen bonding in biological examples.

OBJECTIVE 3

Explain why water properties matter for cells and organisms.

OBJECTIVE 4

Interpret pH as a chemical condition that affects biology.

OBJECTIVE 5

Track atoms through a simple chemical reaction.

BASIC CHEMISTRY

Topic sequence

- Atoms and elements: Atoms form elements, and their electrons shape bonding behavior.
- Bonding and molecular shape: Covalent, ionic, and hydrogen bonds create different molecular properties.
- Water as a biological solvent: Water polarity explains cohesion, temperature buffering, and dissolving ions or polar molecules.
- pH and chemical balance: pH tracks hydrogen ion concentration and affects biological molecules.
- Chemical reactions in cells: Cells depend on chemical reactions that rearrange matter without destroying atoms.

01

Atoms and elements

Atoms form elements, and their electrons shape bonding behavior.

02

Bonding and molecular shape

Covalent, ionic, and hydrogen bonds create different molecular properties.

03

Water as a biological solvent

Water polarity explains cohesion, temperature buffering, and dissolving ions or polar molecules.

04

pH and chemical balance

pH tracks hydrogen ion concentration and affects biological molecules.

05

Chemical reactions in cells

Cells depend on chemical reactions that rearrange matter without destroying atoms.

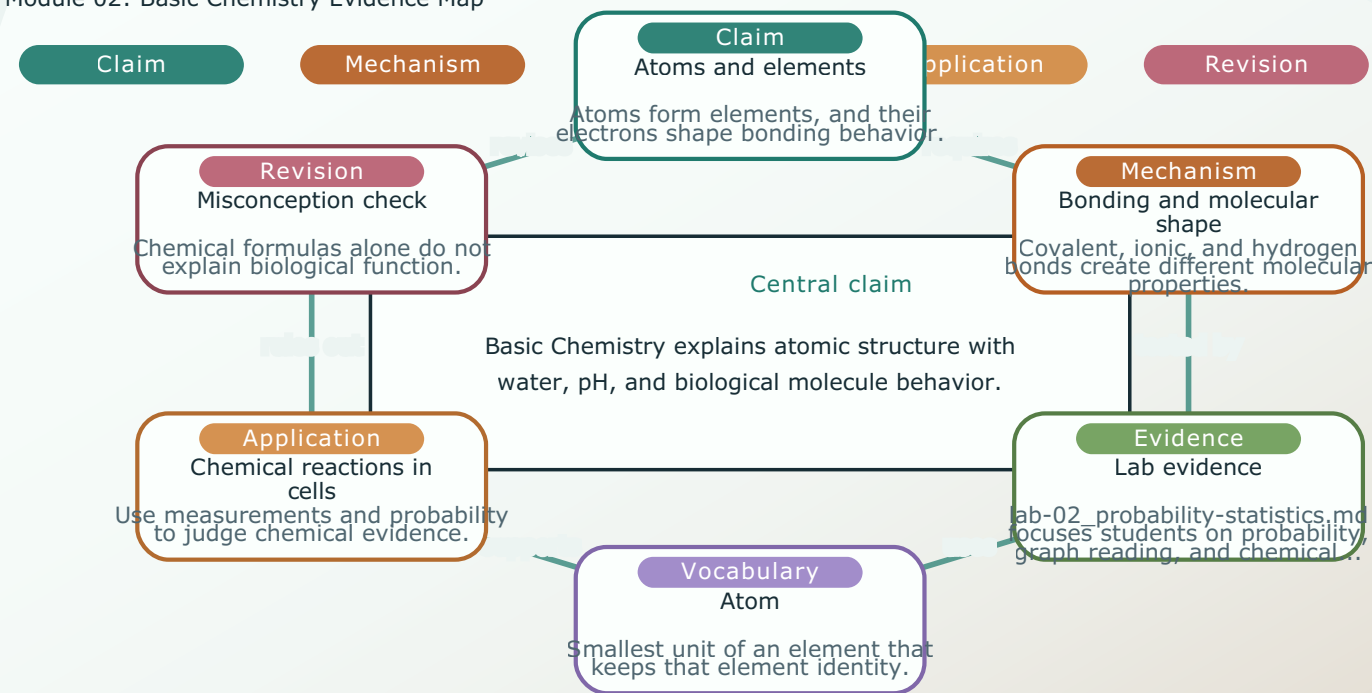
BASIC CHEMISTRY

Module 02: Basic Chemistry Evidence Map

- Module focus: Basic Chemistry
- Central claim: Basic Chemistry explains atomic structure with water, pH, and biological molecule behavior.
- Key nodes to connect: Atoms and elements, Bonding and molecular shape, Lab evidence
- Reasoning move: use 'requires' to explain relationships, not memorize terms.
- Lab connection: lab-02_probability-statistics.md supplies an evidence surface for the map.

Module 02 / Concept Map

Module 02: Basic Chemistry Evidence Map



Linked lab: lab-02_probability-statistics.md / Generated from explicit module.toml visual schema

Teaching note: Ask students to explain one edge in their own words before moving on.

BASIC CHEMISTRY

Module 02: Basic Chemistry Reasoning Sequence

- Module focus: Basic Chemistry
- Trace the model: Atoms and elements -> Bonding and molecular shape -> Water as a biological solvent -> pH and chemical balance
- Inputs become outputs: Atoms and elements, Bonding and molecular shape -> Use atomic structure to explain basic bonding patterns., Distinguish covalent, ionic, and hydrogen bonding in biological examples.
- Feedback check: lab-02_probability-statistics.md evidence checks whether students can explain water as a biological solvent.
- Lab connection: lab-02_probability-statistics.md gives students a process to observe or test.

Module 02 / Process Model

Module 02: Basic Chemistry Reasoning Sequence

Inputs

- Atoms and elements
- Bonding and molecular shape
- Evidence from lab-02_probability-statistics.md

Outputs

- Use atomic structure to explain basic bonding patterns.
- Distinguish covalent, ionic, and hydrogen bonding in biological...
- Track atoms through a simple chemical reaction.

Reasoning sequence

1

Atoms and elements

Atoms form elements, and electrons shape bonding

2

Bonding and molecular shape

Covalent, ionic, and hydrogen bonds create different molecular properties.

3

Water as a biological solvent

Water polarity explains temperature buffering, and dissolving ions or polar.

4

pH and chemical balance

pH tracks hydrogen ion concentration and affects biological molecules.

5

Chemical reactions in cells

Cells depend on chemical reactions that rearrange matter without destroying atoms.

Feedback

lab-02_probability-statistics.md evidence checks whether students can explain water as a biological solvent.; Retrieval questions reveal whether terms...

Constraint

Chemical formulas alone do not explain biological function.; Cells depend on chemical reactions that rearrange matter without destroying atoms.

Linked lab: lab-02_probability-statistics.md

BASIC CHEMISTRY

Terms as evidence handles

- Atom: Smallest unit of an element that keeps that element identity.
- Element: Pure substance defined by proton number.
- Covalent bond: Bond formed when atoms share electrons.
- Ionic bond: Attraction between oppositely charged ions.
- Hydrogen bond: Weak attraction involving polar hydrogen and electronegative atoms.
- Polarity: Uneven charge distribution across a molecule.

ATOM

Smallest unit of an element that keeps that element identity.

ELEMENT

Pure substance defined by proton number.

COVALENT BOND

Bond formed when atoms share electrons.

IONIC BOND

Attraction between oppositely charged ions.

HYDROGEN BOND

Weak attraction involving polar hydrogen and electronegative atoms.

POLARITY

Uneven charge distribution across a molecule.

BASIC CHEMISTRY

Lab connection

- Lab file: lab-02_probability-statistics.md
- Lab evidence should test or illustrate:
Water as a biological solvent
- Students should leave with a concrete observation, comparison, or model tied to the module claim.

QUESTION

Atoms and elements

EVIDENCE

lab-02_probability-statistics.md

CLAIM

Use atomic structure to explain basic bonding patterns.

BASIC CHEMISTRY

Contrast check

- Do not stop at: Atoms form elements, and their electrons shape bonding behavior.
- Stronger explanation: Cells depend on chemical reactions that rearrange matter without destroying atoms.
- The goal is to move from a named idea to a testable biological explanation.

SURFACE

Atoms form elements, and their electrons shape bonding behavior.

DEEPER

Cells depend on chemical reactions that rearrange matter without destroying atoms.

BASIC CHEMISTRY

Module 02: Basic Chemistry Retrieval and Lab Check

- Module focus: Basic Chemistry
- Retrieval prompt: How do protons, neutrons, and electrons differ?
- Answer check: Use atomic structure to explain basic bonding patterns.
- Required terms: Atom, Element, Covalent bond
- Lab connection: lab-02_probability-statistics.md; lab-02_probability-statistics.md supplies evidence for probability, graph reading, and chemical evidence claims.

Module 02 / Retrieval Card

Module 02: Basic Chemistry Retrieval and
Lab Check

Routine

Cover notes -> answer aloud
-> cite evidence -> revise

1 Cover notes

2 Answer aloud

3 Cite evidence

4 Revise

How do protons, neutrons, and electrons differ?

Check: Use atomic structure to explain basic bonding patterns.

Why do electrons matter for chemical bonding?

Check: Distinguish covalent, ionic, and hydrogen bonding in biological examples.

What distinguishes covalent, ionic, and hydrogen bonds?

Check: Explain why water properties matter for cells and organisms.

Why does water dissolve salt but not all nonpolar molecules?

Check: Interpret pH as a chemical condition that affects biology.

Terms to use

Atom, Element, Covalent bond, Ionic bond, Hydrogen bond

Lab connection

lab-02_probability-statistics.md supplies evidence for probability, graph reading, and chemical evidence claims.

Intro biology routine: claim, evidence, reasoning, revision.

Teaching note: Pause here. Students answer first without notes, then compare to the checks.

BASIC CHEMISTRY

Practice quiz bridge

- 1. Table salt (NaCl) forms when a sodium atom gives up an electron to a chlorine atom. What kind of bond holds the result together?
- 2. Why can water dissolve so many substances important to cells?
- 3. A solution changes from pH 7 to pH 4. This means it became:
- 4. Blood keeps a nearly steady pH even when acids enter it. What best explains this stability?

QUESTION 1**Key: B**

Table salt (NaCl) forms when a sodium atom gives up an electron to a chlorine atom. What kind of bond holds the result together?

QUESTION 2**Key: A**

Why can water dissolve so many substances important to cells?

QUESTION 3**Key: D**

A solution changes from pH 7 to pH 4. This means it became:

QUESTION 4**Key: C**

Blood keeps a nearly steady pH even when acids enter it. What best explains this stability?

BASIC CHEMISTRY

Synthesis and exit ticket

- Exit claim: explain atoms and elements using evidence from lab-02_probability-statistics.md.
- Must include: Atom, Element, Covalent bond.
- Revision prompt: what evidence would change your explanation?

CLAIM

Atoms and elements

EVIDENCE

lab-02_probability-statistics.md

REVISION

What would change your mind?